

## Dec 1, 2025 (Due: 08:00 Dec 8, 2025)

1. Let  $A \in \mathbb{C}^{n \times n}$  be Hermitian and positive definite, and  $b \in \mathbb{C}^n$ . Suppose that  $\mathcal{V}$  is a subspace of  $\mathbb{C}^n$ . Show that

$$x_0 = \arg \min_{x \in \mathcal{V}} \|x - A^{-1}b\|_A$$

if and only if  $b - Ax_0 \in \mathcal{V}^\perp$ . (The orthogonal complement is defined using the standard inner product.)

2. Let  $A \in \mathbb{C}^{n \times n}$  be Hermitian and positive definite, and  $b \in \mathbb{C}^n$ . Suppose that  $A$  has  $m$  distinct eigenvalues. Show that the CG method for solving  $Ax = b$  converges in at most  $m$  iterations.

3. Suppose you are given a matrix  $A \in \mathbb{C}^{m \times n}$  with full column rank, and a vector  $b \in \mathbb{C}^m$ . To solve the least squares problem  $\min_x \|Ax - b\|_2$ , the CG method can be applied to solve the normal equation  $A^*Ax = A^*b$ . Provide a detailed iterative scheme for this method. Make sure that the operation  $v \mapsto A^*Av$  is avoided.

4. Implement the CG method and test it on a randomly generated positive definite linear system with 1000 unknowns. Visualize the convergence history.

Make a record of  $R = [r_0, r_1, \dots]$  and  $P = [p_0, p_1, \dots]$ . Normalize these vectors properly, and visualize the *componentwise* losses of orthogonality  $|R^*R - I|$  and  $|P^*AP - I|$ .

5. Use the CG method to solve the 2D Laplace equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

over the unit square  $[0, 1]^2$  with boundary conditions

$$u(0, y) = u(1, y) = u(x, 1) = 0, \quad u(x, 0) = \sin(\pi x).$$

Visualize the solution and the convergence history.

(optional) Use ICchol-based preconditioning to accelerate the convergence of CG. (ICchol means incomplete Cholesky factorization.)

6. (optional) Let  $p(\lambda) = \sum_{k=0}^n a_k \lambda^k$  be a real polynomial such that

$$\max_{-1 \leq \lambda \leq 1} |p(\lambda)| \leq 1.$$

Show that  $|a_n| \leq 2^{n-1}$ .

7. (optional) Let  $p(\lambda)$  be a real polynomial such that  $\deg p(\lambda) \leq n$  and

$$\max_{-1 \leq \lambda \leq 1} |p(\lambda)| \leq 1.$$

Show that  $|p(\mu)| \leq |T_n(\mu)|$  for any  $\mu \in \mathbb{R} \setminus [-1, 1]$ .